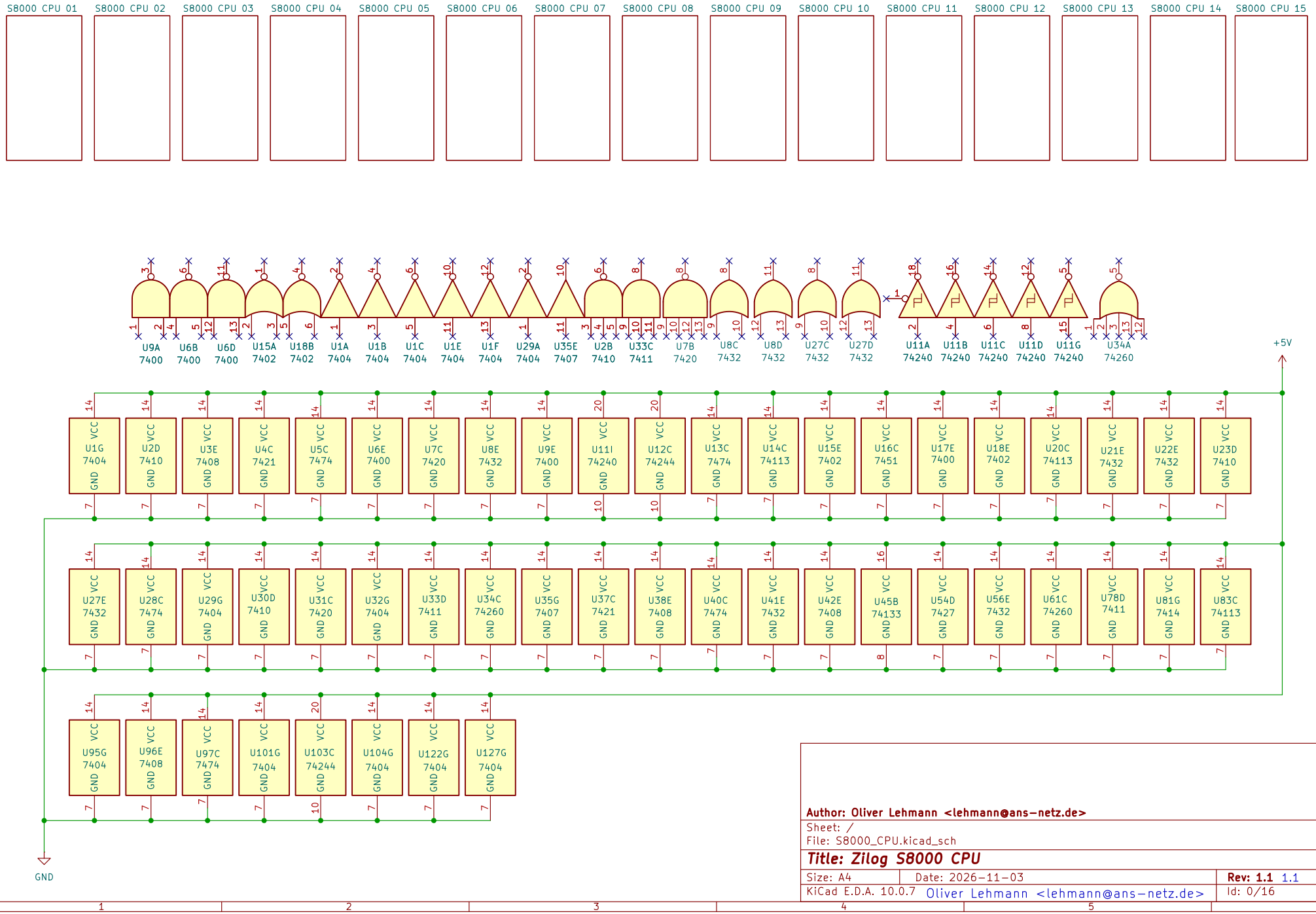

S8000 CPU 01
S8000 CPU 02
S8000 CPU 03
S8000 CPU 04
S8000 CPU 05
S8000 CPU 06
S8000 CPU 07
S8000 CPU 08
S8000 CPU 09
S8000 CPU 10
S8000 CPU 11
S8000 CPU 12
S8000 CPU 13
S8000 CPU 14
S8000 CPU 15
U9A 7400
U6B 7400
U6D 7400
U15A 7402
U18B 7402
U1A 7404
U1B 7404
U1C 7404
U1E 7404
U1F 7404
U29A 7404
U35E 7407
U2B 7410
U33C 7411
U7B 7420
U8C 7432
U8D 7432
U27C 7432
U27D 7432
U11A 74240
U11B 74240
U11C 74240
U11D 74240
U11G 74240
U34A 74260
+5V
U1G 7404
U2D 7410
U3E 7408
U4C 7421
U5C 7474
U6E 7400
U7C 7420
U8E 7432
U9E 7400
U11I 74240
U12C 74244
U13C 7474
U14C 74113
U15E 7402
U16C 7451
U17E 7400
U18E 7402
U20C 74113
U21E 7432
U22E 7432
U23D 7410
U27E 7432
U28C 7474
U29G 7404
U30D 7410
U31C 7420
U32G 7404
U33D 7411
U34C 74260
U35G 7407
U37C 7421
U38E 7408
U40C 7474
U41E 7432
U42E 7408
U45B 74133
U54D 7427
U56E 7432
U61C 74260
U78D 7411
U81G 7414
U83C 74113
U95G 7404
U96E 7408
U97C 7474
U101G 7404
U103C 74244
U104G 7404
U122G 7404
U127G 7404
GND
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /
File: S8000_CPU.kicad_sch
Title: Zilog S8000 CPU
Size: A4
Date: 2026-11-03
Rev: 1.1 1.1
KiCad E.D.A. 10.0.7 Oliver Lehmann <lehmann@ans-netz.de>
Id: 0/16

I.C.	+5V (PIN)	GND	−5V (PIN)	DESIGNATIONS	SPARE GATES
74S00	14	7		U6,9,17	U9 − A,D, U6 − B,D
74S02	14	7		U15	U15 − A
74LS02	14	7		U18	U18 − B
74S04	14	7		U1,29,95	U1 − A−C,E,F, U29 − A
74LS04	14	7		U32,101,104,122, 127	
7407	14	7		U35	U35 − E
74S08	14	7		U3,96	
74LS08	14	7		U38,42	
74S10	14	7		U2,23,30	U2 − B
74S11	14	7		U78	
74LS11	14	7		U33	U33 − C
74LS14	14	7		U81	
74S20	14	7		U7,31	U7 − B
74LS21	14	7		U4,37	
74LS27	14	7		U54	
74S32	14	7		U21	
74LS32	14	7		U8,22,27,41	U8 − C,D, U27 − C,D
74S51	14	7		U16	
74S74	14	7		U5,13,97	
74LS74	14	7		U28,40	
74S85	14	7		U52,53	
74S113 1)	14	7		U14,20,83	
74LS133	16	8		U45	
74LS138 2)	16	8		U24−26,39,55	
74LS175	16	8		U36	

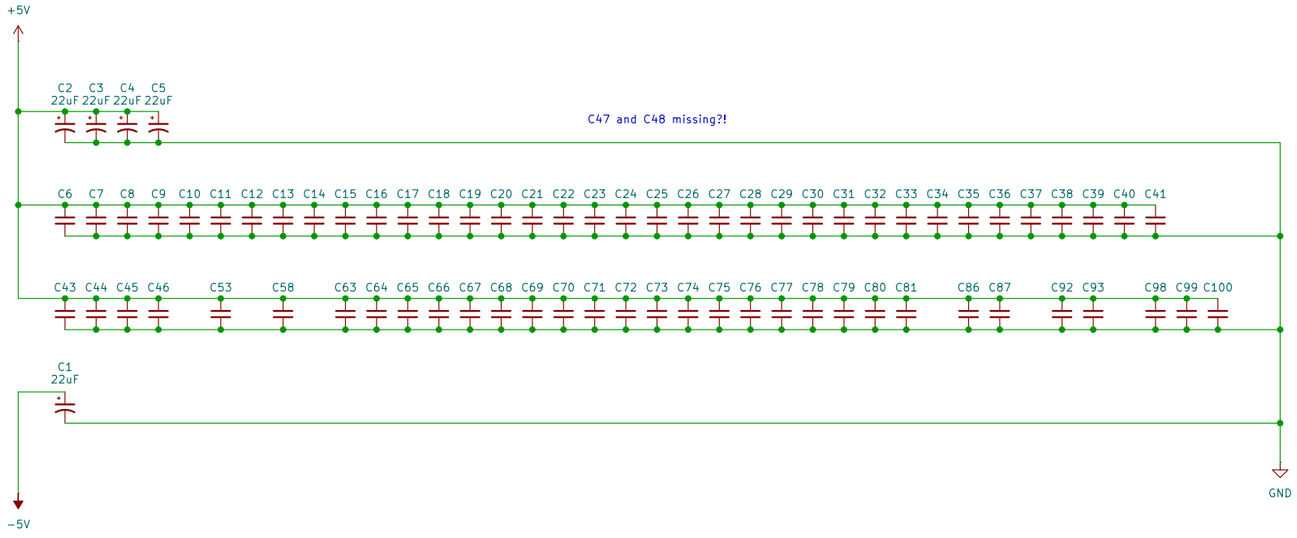
1) U83 is a LS on my board

2) U25 and U26 are both S on my board

I.C. TYPE	+5V (PIN)	GND	−5V (PIN)	DESIGNATIONS	SPARE GATES
74S240	20	10		U11,68,109	
74S241	20	10		U99	
74LS244	20	10		U12,51,67,73,103, 108,110,112,114− 118	
74LS245	20	10		U46,63.111,113	
74S260	14	7		U34	U34 − A
74LS260	14	7		U61	~~U61 − B~~
74S373	20	10		U47,62,71,72	
74LS374	20	10		U48,49,50,65,66	
26LS30	1	5	8	U100,102,105,124, 126,128	
26LS32	16	8		U106,107,120,121, 123,125	
93S16	16	8		U98	
Z80B−CTC	25	5		U43,44,89	
2114(p)−3	18	9		U57−60	
Z8001A−CPU	11	36		U64	
2732	24	12		U74−77	
Z80CLK DRV	1	5		U79,80	
2.4576 MHz	14	7		U84	
44.440 MHz	14	7		U85	
Z8010A MMU	11	35		U86,87,88	
Z80B−SIO/2	9	31		U90−93	
Z80B−PIO	26	11		U94	

REF DESIGNATION	
LAST USED	NOT USED
C100	–––
CR1	–––
DS4	–––
R20	–––
U128	–––

NOTES: UNLESS OTHERWISE SPECIFIED
1. ALL RESISTORS ARE 1Ka, 1/4W, 5%
2. CAPACITOR VALUES ARE IN MICROFARADS.
3. ALPHANUMERIC DENOTES GATES

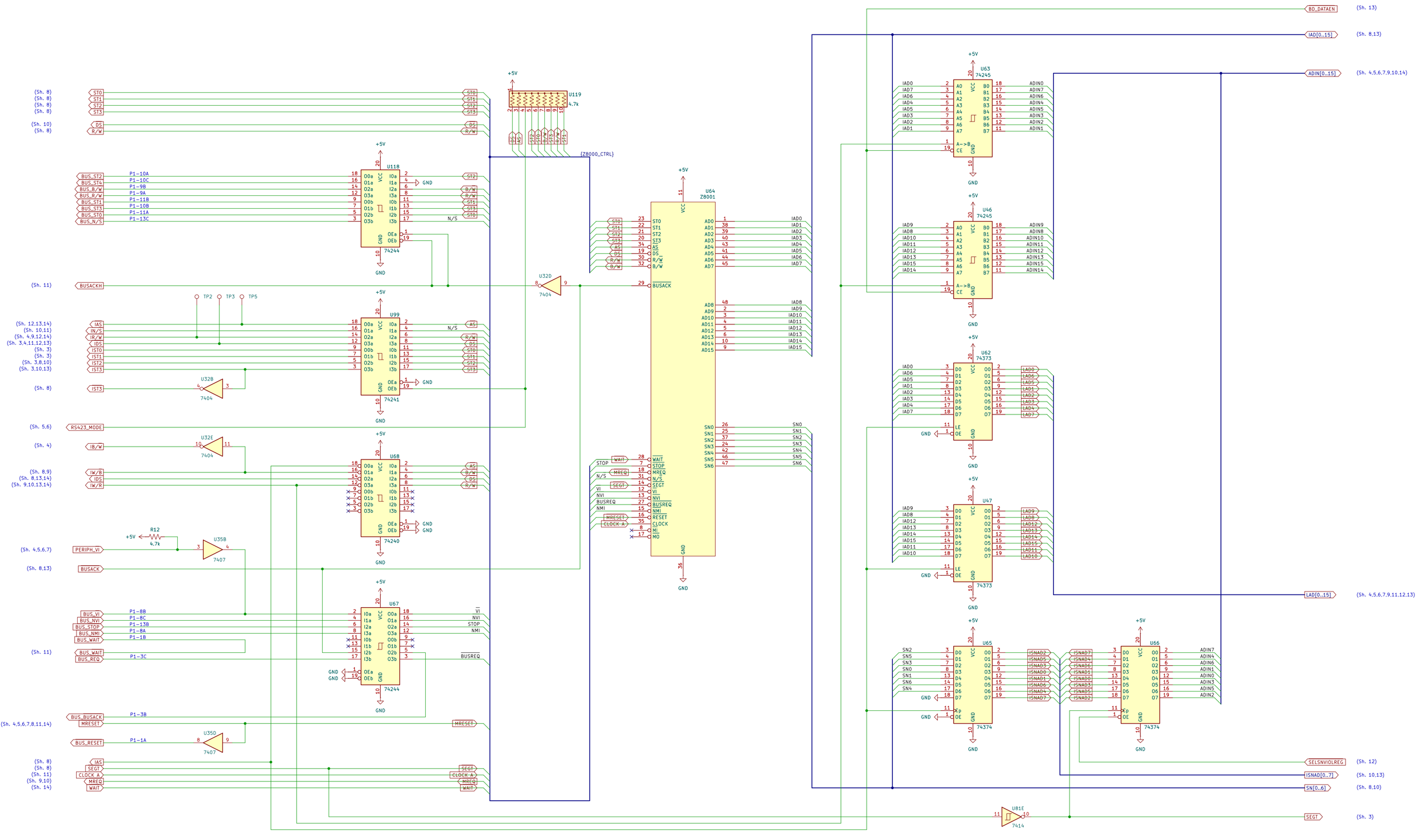


Author: Oliver Lehmann <lehmann@ans−netz.de>

Sheet: /S8000 CPU 01/
File: S8000_CPU_01.kicad_sch

Title: Zilog S8000 CPU

Size: A3	Date: 2026−11−03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 1/16



Z8001A CPU
Segment Violation Register U66
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 02/
File: S8000_CPU_02.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 2/16

(Sh. 2) IAD[0..15]

ADIN[0..15] (Sh. 13)

U48 74374
U50 74374
U49 74374

IAD7 IAD6 IAD5 IAD4 IAD1 IAD2 IAD3 IAD0 → U48 D0..D7
U48 O0..O7 → ADIN7 ADIN6 ADIN5 ADIN4 ADIN3 ADIN2 ADIN1 ADIN0
U50 D0..D7 ← IAD7 IAD6 IAD5 IAD4 IAD1 IAD2 IAD3 IAD0
U50 O0..O7 → ADIN7 ADIN6 ADIN5 ADIN4 ADIN3 ADIN2 ADIN1 ADIN0
U48 OE, U50 OE ← GND
U49 D0..D7 ← IAD7 IAD6 IAD5 IAD4 IAD1 IAD2 IAD3 IAD0
U49 O0..O7 → ADIN7 ADIN6 ADIN5 ADIN4 ADIN1 ADIN2 ADIN3 ADIN0

(Sh. 12) IF1LOBYTE → U48 OE, U50 OE
(Sh. 2) SEGT
(Sh. 12) TRPLOBYTE
SEGT, TRPLOBYTE → U49 OE

U27A 7432
IAD0 → U27A pin 1
U27A pin 2 ← U49 OE

U25 74138
(Sh. 2) IST0 → U25 A0
(Sh. 2) IST1 → U25 A1
(Sh. 2) IST2 → U25 A2
(Sh. 2) IST3 → R4 1k → +5V; U25 E3
U25 E2 ← U27A pin 3
U25 E1 ← GND
U25 O0 → I/O_REF (Sh. 11.12)
U25 O1 → SEGSTACK (Sh. 8)
U25 O2 → VIACK (Sh. 12)

R2 1k, R3 1k → +5V

U4A 7421
U25 O3 → U4A pin 1
U25 O4 → U4A pin 2
U25 O5 → U4A pin 4
U4A pin 5 ← U25 O6
U4A pin 6 → MMU/EPUREF (Sh. 13)

U26 74138
U25 A0 → U26 A0
U25 A1 → U26 A1
U25 A2 → U26 A2
U26 O0 → STACKMEMREQ (Sh. 4)
U26 O5 → INSTFETCH1STWD (Sh. 8)
U26 O6 → EPUXFER (Sh. 11)
U26 E3 ← U25 O6
U26 E2, E1 ← GND

P2_NMI_SWITCH_NO P2-26C → U9C 7400
P2_NMI_SWITCH_NC P2-25C → U9D 7400
(Sh. 4) MANUAL_NMI → U9C, U9D

(Sh. 4) LATCHECCERROR → U54B 7427
(Sh. 10.11) POWERFAIL → U54B 7427
U54B 7427 → U35C 7407 → TP6 → P1-8A BUS_NMI

UBA 7432
U25 O4 → UBA pin 1
U25 O6 → UBA pin 2
UBA pin 3 → ONBDNMI (Sh. 12)

U8B 7432
(Sh. 2) IDS → U8B pin 5
U8B pin 6 → ENAONBDNMI (Sh. 10)

External Violation Registers (U48,49,50)
VIACK- generation
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 03/
File: S8000_CPU_03.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 3/16

(Sh. 2) ADIN[0..15]

+5V

U44
Z80 CTC

ADIN0 25 D0
ADIN1 26 D1
ADIN2 27 D2
ADIN3 28 D3
ADIN4 1 D4
ADIN5 2 D5
ADIN6 3 D6
ADIN7 4 D7

(Sh. 12) CTC 0 CE → 16 CE
18 CS0
19 CS1
14 M1
10 IOREQ
6 RD
12 INT
17 RESET
CTC 0

U41A
7432
(Sh. 2) IDS → 1
(Sh. 3) STACKMEMREQ → 2
3 → 15 CLK

23 CLK/TRG0
22 CLK/TRG1
21 CLK/TRG2
20 CLK/TRG3

+5V → 13 IEI
ZC/TO0 7
ZC/TO1 8
ZC/TO2 9
IEO 11

TP8

BAUD 0 (Sh. 5)
BAUD 1 (Sh. 5)
BAUD 2 (Sh. 5)
CTC 0 IEO (Sh. 14)

24 VCC
5 GND
GND

+5V

U43
Z80 CTC

ADIN0 25 D0
ADIN1 26 D1
ADIN2 27 D2
ADIN3 28 D3
ADIN4 1 D4
ADIN5 2 D5
ADIN6 3 D6
ADIN7 4 D7

TP10

(Sh. 12) CTC 1 CE → 16 CE
(Sh. 2) LAD1 → 18 CS0
(Sh. 2) LAD2 → 19 CS1
(Sh. 12) M1 → 14 M1
(Sh. 12) IOREQ → 10 IOREQ
(Sh. 12) RD → 6 RD
(Sh. 2) PERIPH_WI → 12 INT
(Sh. 2) MRESET → 17 RESET
CTC 1

(Sh. 11) CLOCK_B → 15 CLK
(Sh. 11) BAUD_CLK → 23 CLK/TRG0
22 CLK/TRG1
21 CLK/TRG2
20 CLK/TRG3

13 IEI
ZC/TO0 7
ZC/TO1 8
ZC/TO2 9
IEO 11

BAUD 3 (Sh. 5)
BAUD 4 (Sh. 5)
BAUD 5 (Sh. 5)
CTC 1 IEO (Sh. 7.14)

24 VCC
5 GND
GND

U33A
7411
(Sh. 2) IR/W → 1
(Sh. 9) LADD → 2
(Sh. 2) IB/W → 13
12

ENA BYTE SWAP (Sh. 13)

U32A
7404
1 → 2
ENA BYTE SWAP (Sh. 13)

(Sh. 14) SYS_DS → 4

GND → 3 J
1 PRE
5 Q
(Sh. 14) CLRPARITY → 1 K
2 CLR
6 Q̄

U14A
74113

(Sh. 14) BUS_ME → P1-12A → U81A 1 → 2
7414

LATCHECCERROR (Sh. 3.10,11)

Serial Input/Output: Baud Rate Generators
Byte Swap Buffer: create the input buffer enable signal
Parity Error Checking: parity error flip-flop
Author: Oliver Lehmann <lehmann@ans-netz.de>

Sheet: /S8000 CPU 04/
File: S8000_CPU_04.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 4/16

(Sh. 2) ADIN[0..15]

(Sh. 12) SIO 0 CE

(Sh. 7.14) CTC 2 IEO

(Sh. 4) BAUD 0

(Sh. 4) BAUD 1

+5V

U93
Z80 SIO/2

ADIN0 40 D0
ADIN1 1 D1
ADIN2 39 D2
ADIN3 2 D3
ADIN4 38 D4
ADIN5 3 D5
ADIN6 37 D6
ADIN7 4 D7

34 B/A
33 C/D

35 CE
21 RESET
8 M1
36 IORQ
32 RD

20 CLK

5 INT
6 IEI
7 IEO

13 RxCA
14 TxCA

28 RxCB
27 TxCB

9 VCC
31 GND

W/RDYA 10
DTRA 16
RTSA 17
TxDA 15
DCDA 19
CTSA 18
RxDA 12
SYNCA 11
W/RDYB 30
DTRB 25
RTSB 24
TxDB 26
DCDB 22
CTSB 23
RxDB 29

SIO 0

GND

U127A 1 2 7404
U127E 11 10 7404
U127D 9 8 7404
U127C 5 6 7404
U127B 3 4 7404
U127F 13 12 7404

U128A 2 15 26LS30
U128C 6 11 26LS30
U128D 7 10 26LS30
U128B 3 14 26LS30

C94 C
C96 C
C97 C
C91 C
C88 C
C95 C

U107D 13 15 26LS32
U107A 3 1 26LS32
U107C 11 9 26LS32
U106D 13 15 26LS32
U106C 11 9 26LS32
U107B 5 7 26LS32

U126D 7 10 26LS30
U126A 2 15 26LS30

P2 – 4A P2_SIO_DSR0
P2 – 2B P2_SIO_CTS0
P2 – 1B P2_SIO_RXD0
P2 – 3B P2_SIO_DTR0
P2 – 3A P2_SIO_RTS0
P2 – 2A P2_SIO_TXD0
P2 – 1A P2_SIO_TXRTN0

P2 – 7A P2_SIO_DSR1
P2 – 5B P2_SIO_CTS1
P2 – 4B P2_SIO_RXD1
P2 – 6B P2_SIO_DTR1
P2 – 6A P2_SIO_RTS1
P2 – 5A P2_SIO_TXD1
P2 – 14C P2_SIO_TXRTN1

SIO 0 IEO (Sh. 14)

(Sh. 2) LAD1
(Sh. 2) LAD2

(Sh. 12) SIO 1 CE
(Sh. 12) MRESET
(Sh. 12) M1
(Sh. 12) IOREQ
(Sh. 12) RD

(Sh. 11) CLOCK B

(Sh. 2) PERIPH_VI

(Sh. 4) BAUD 2

(Sh. 4) BAUD 3

+5V

U92
Z80 SIO/2

ADIN0 40 D0
ADIN1 1 D1
ADIN2 39 D2
ADIN3 2 D3
ADIN4 38 D4
ADIN5 3 D5
ADIN6 37 D6
ADIN7 4 D7

34 B/A
33 C/D

35 CE
21 RESET
8 M1
36 IORQ
32 RD

20 CLK

5 INT
6 IEI
7 IEO

13 RxCA
14 TxCA

28 RxCB
27 TxCB

9 VCC
31 GND

W/RDYA 10
DTRA 16
RTSA 17
TxDA 15
DCDA 19
CTSA 18
RxDA 12
SYNCA 11
W/RDYB 30
DTRB 25
RTSB 24
TxDB 26
DCDB 22
CTSB 23
RxDB 29

SIO 1

GND

U104B 3 4 7404
U104A 1 2 7404
U104E 11 10 7404
U104D 9 8 7404
U104C 5 6 7404
U104F 13 12 7404

U126B 3 14 26LS30
U105A 2 15 26LS30
U126C 6 11 26LS30
U125D 13 15 26LS32
U106B 5 7 26LS32
U106A 3 1 26LS32

C89 C
C59 C
C90 C

U105D 7 10 26LS30
U105B 3 14 26LS30
U105C 6 11 26LS30
U125A 3 1 26LS32
U125B 5 7 26LS32
U125C 11 9 26LS32

C62 C
C60 C
C61 C

P2 – 10A P2_SIO_DSR2
P2 – 8B P2_SIO_CTS2
P2 – 7B P2_SIO_RXD2
P2 – 9B P2_SIO_DTR2
P2 – 9A P2_SIO_RTS2
P2 – 8A P2_SIO_TXD2
P2 – 18C P2_SIO_TXRTN2

P2 – 13A P2_SIO_DSR3
P2 – 11B P2_SIO_CTS3
P2 – 10B P2_SIO_RXD3
P2 – 12B P2_SIO_DTR3
P2 – 12A P2_SIO_RTS3
P2 – 11A P2_SIO_TXD3
P2 – 26A P2_SIO_TXRTN3

SIO 1 IEO (Sh. 14)

−5V +5V

U105E 26LS30
U126E 26LS30
U128E 26LS30
U106E 26LS32
U107E 26LS32
U125E 26LS32

(Sh. 2) RS423_MODE

GND

Serial Input/Output: SIO channels 0 through 3
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 05/
File: S8000_CPU_05.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 5/16

(Sh. 2) ADIN[0..15]

(Sh. 12) SIO 2 CE

(Sh. 14) SIO 2 IEI

(Sh. 4) BAUD 4

(Sh. 4) BAUD 5

+5V

U91
Z80 SIO/2

ADIN0 40 D0
ADIN1 1 D1
ADIN2 39 D2
ADIN3 2 D3
ADIN4 38 D4
ADIN5 3 D5
ADIN6 37 D6
ADIN7 4 D7

9 VCC

34 B/A
33 C/D

35 CE
21 RESET
8 M1
36 IORQ
32 RD

20 CLK

5 INT
6 IEI
7 IEO

13 RxCA
14 TxCA

28 RxCB
27 TxCB

10 W/RDYA
16 DTRA
17 RTSA
15 TxDA
19 DCDA
18 CTSA
12 RxDA
11 SYNCA
30 W/RDYB
25 DTRB
24 RTSB
26 TxDB
22 DCDB
23 CTSB
29 RxDB

SIO 2

31 GND

GND

U122F 13 12 7404
U122C 5 6 7404
U122D 9 8 7404

U124A 2 15 26LS30
U124C 6 11 26LS30
U124D 7 10 26LS30

C82 C
C84 C
C85 C

U123A 3 2 1 26LS32
U123C 11 10 9 26LS32
U123D 13 14 15 26LS32

P2 – 16A P2_SIO_DSR4
P2 – 14B P2_SIO_CTS4
P2 – 13B P2_SIO_RXD4
P2 – 15B P2_SIO_DTR4
P2 – 15A P2_SIO_RTS4
P2 – 14A P2_SIO_TXD4
P2 – 27A P2_SIO_TXRTN4

U122B 3 4 7404
U122A 1 2 7404
U122E 11 10 7404

U102B 3 13 14 26LS30
U102D 7 9 10 26LS30
U124B 3 13 14 26LS30

C55 C
C57 C
C83 C

U121D 13 14 15 26LS32
U121C 11 10 9 26LS32
U123B 5 6 7 26LS32

P2 – 19A P2_SIO_DSR5
P2 – 17B P2_SIO_CTS5
P2 – 16B P2_SIO_RXD5
P2 – 18B P2_SIO_DTR5
P2 – 18A P2_SIO_RTS5
P2 – 17A P2_SIO_TXD5
P2 – 25B P2_SIO_TXRTN5

SIO 2 IEO (Sh. 14)

(Sh. 2) LAD1
(Sh. 2) LAD2

(Sh. 12) SIO 3 CE
(Sh. 12) MRESET
(Sh. 12) M1
(Sh. 12) IOREQ
(Sh. 12) RD

(Sh. 11) CLOCK B

(Sh. 2) PERIPH_VI
(Sh. 14) SIO 3 IEI

(Sh. 7) BAUD 6

(Sh. 7) BAUD 7

+5V

U90
Z80 SIO/2

ADIN0 40 D0
ADIN1 1 D1
ADIN2 39 D2
ADIN3 2 D3
ADIN4 38 D4
ADIN5 3 D5
ADIN6 37 D6
ADIN7 4 D7

9 VCC

34 B/A
33 C/D

35 CE
21 RESET
8 M1
36 IORQ
32 RD

20 CLK

5 INT
6 IEI
7 IEO

13 RxCA
14 TxCA

28 RxCB
27 TxCB

10 W/RDYA
16 DTRA
17 RTSA
15 TxDA
19 DCDA
18 CTSA
12 RxDA
11 SYNCA
30 W/RDYB
25 DTRB
24 RTSB
26 TxDB
22 DCDB
23 CTSB
29 RxDB

SIO 3

31 GND

GND

U101C 5 6 7404
U101F 13 12 7404
U101E 11 10 7404

U100C 6 11 26LS30
U102A 2 15 26LS30
U102C 6 11 26LS30

C51 C
C54 C
C56 C

U120D 13 14 15 26LS32
U121B 5 6 7 26LS32
U121A 3 2 1 26LS32

P2 – 22A P2_SIO_DSR6
P2 – 20B P2_SIO_CTS6
P2 – 19B P2_SIO_RXD6
P2 – 21B P2_SIO_DTR6
P2 – 21A P2_SIO_RTS6
P2 – 20A P2_SIO_TXD6
P2 – 26B P2_SIO_TXRTN6

U101A 1 2 7404
U101B 3 4 7404
U101D 9 8 7404

U100A 2 15 26LS30
U100B 3 13 14 26LS30
U100D 7 9 10 26LS30

C49 C
C50 C
C52 C

U120A 3 2 1 26LS32
U120B 5 6 7 26LS32
U120C 11 10 9 26LS32

P2 – 25A P2_SIO_DSR7
P2 – 23B P2_SIO_CTS7
P2 – 22B P2_SIO_RXD7
P2 – 24B P2_SIO_DTR7
P2 – 24A P2_SIO_RTS7
P2 – 23A P2_SIO_TXD7
P2 – 27B P2_SIO_TXRTN7

SIO 3 IEO (Sh. 14)

−5V +5V

U100E 26LS30 4 MODE VEE 8 5 GND
U102E 26LS30 4 MODE VEE 8 5 GND
U124E 26LS30 4 MODE VEE 8 5 GND
U120E 26LS32 4 VCC 16 G 8 GND
U121E 26LS32 4 VCC 16 G 8 GND
U123E 26LS32 4 VCC 16 G 8 GND

(Sh. 2) RS423_MODE

GND

Serial Input/Output: SIO channels 4 through 7
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 06/
File: S8000_CPU_06.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 6/16

(Sh. 2) ADIN[0..15]

(Sh. 12) CTC 2 CE

(Sh. 2) MRESET

(Sh. 11) BAUD CLK

(Sh. 4) CTC 1 IEO

U89
Z80 CTC
CTC 0

+5V

ADIN0 ADIN1 ADIN2 ADIN3 ADIN4 ADIN5 ADIN6 ADIN7

CE CS0 CS1 M1 IOREQ RD INT RESET

CLK CLK/TRG0 CLK/TRG1 CLK/TRG2 CLK/TRG3 IEI

ZC/TO0 ZC/TO1 ZC/TO2 IEO

VCC GND

BAUD 6 (Sh. 6)
BAUD 7 (Sh. 6)
CTC 2 IEO (Sh. 5)

A[0..7]

B[0..7]

U94
Z80 PIO

+5V

ADIN0 ADIN1 ADIN2 ADIN3 ADIN4 ADIN5 ADIN6 ADIN7

B/A C/D CE M1 IORQ RD CLK INT IEI IEO

PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7

PB0 PB1 PB2 PB3 PB4 PB5 PB6 PB7

ARDY ASTB BRDY BSTB

VCC GND

(Sh. 2) LAD1
(Sh. 2) LAD2
(Sh. 12) PIO CE
(Sh. 12) M1
(Sh. 12) IOREQ
(Sh. 12) RD
(Sh. 11) CLOCK B
(Sh. 2) PERIPH_VI
(Sh. 14) PIO_0_IEI

U42B
7408

Data Products Interface
E16 E17
E17_DEF
E18
Centronics Interface

U11E
74240

U103A
74244

U103B
74244

I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
OEa OEb

R20 1k +5V
R19 1k +5V
R5 1k +5V

P2 – 12C P2_PIO_DSTRB/DSTRB
P2 – 13C P2_PIO_N.U./INPUT PRIME
P2 – 20C P2_PIO_IFVALID/FAULT
P2 – 21C P2_PIO_ON-LINE/SELECT
P2 – 17C P2_PIO_BUSY

U108
74244

B0 B1 B2 B3 B4 B5 B6 B7

P2 – 1C P2_PIO_DATA0
P2 – 2C P2_PIO_DATA1
P2 – 3C P2_PIO_DATA2
P2 – 5C P2_PIO_DATA3
P2 – 6C P2_PIO_DATA4
P2 – 8C P2_PIO_DATA5
P2 – 9C P2_PIO_DATA6
P2 – 10C P2_PIO_DATA7

Data Products Interface
E14 E15
E14_DEF
E13
Centronics Interface

U11F
74240

U128
74244

R6 1k +5V

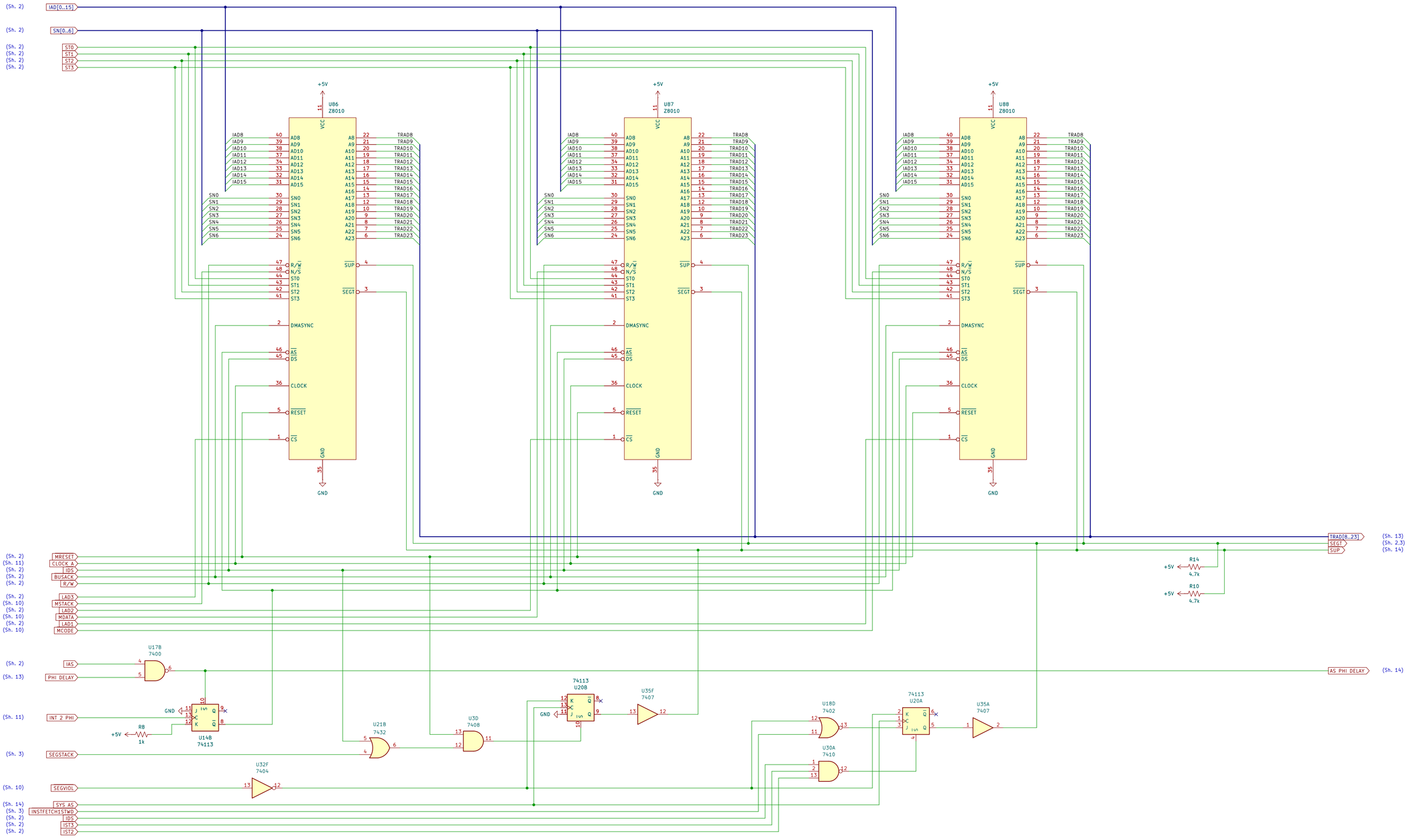
P2 – 16C P2_PIO_D.D./ACKNOWLEDGE

PIO_0_IEO (Sh. 14)

Serial Input/Output: Baud Rate Generators
Parallel I/O Ports
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 07/
File: S8000_CPU_07.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 7/16



Memory Management Units

Author: Oliver Lehmann <lehmann@ans-netz.de>

Sheet: /S8000 CPU 08/

File: S8000_CPU_08.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 8/16

(Sh. 2) ADIN[0..15]

(Sh. 2) LAD[0..15]

U74 2732 (1L) — +5V, VCC, GND; LAD1–LAD12 → A0–A11; D0–D7 → ADIN0–ADIN7; OE/VPP, CE → GND

U75 2732 (2L) — +5V, VCC, GND; LAD1–LAD12 → A0–A11; D0–D7 → ADIN0–ADIN7; OE/VPP, CE → GND

U59 2114 — +5V, VCC, GND; LAD1–LAD10 → A0–A9; DQ1–DQ4 → ADIN8–ADIN11; WE, CS → GND

U57 2114 — +5V, VCC, GND; LAD1–LAD10 → A0–A9; DQ1–DQ4 → ADIN0–ADIN3; WE, CS → GND

U76 2732 (1H) — +5V, VCC, GND; LAD1–LAD12 → A0–A11; D0–D7 → ADIN8–ADIN15; OE/VPP, CE → GND

U77 2732 (2H) — +5V, VCC, GND; LAD1–LAD12 → A0–A11; D0–D7 → ADIN8–ADIN15; OE/VPP, CE → GND

U60 2114 — +5V, VCC, GND; LAD1–LAD10 → A0–A9; DQ1–DQ4 → ADIN12–ADIN15; WE, CS → GND

U58 2114 — +5V, VCC, GND; LAD1–LAD10 → A0–A9; DQ1–DQ4 → ADIN4–ADIN7; WE, CS → GND

U56A 7432

(Sh. 10) BDMEMA2K

(Sh. 2) IW/R

U56D 7432

(Sh. 10) BDMEMB2K

TP9

(Sh. 2) MREQ

(Sh. 2) IR/W

U27B 7432

(Sh. 4) LAD0

U81F 7414

(Sh. 2) LAD0

U54C 7427

U56C 7432

MEMC2KHI

U54A 7427

(Sh. 2) IW/B

U56B 7432

MEMC2KLO

(Sh. 10) BDMEMC2K

Read-Only Memory
Read/Write Memory
Author: Oliver Lehmann <lehmann@ans-netz.de>

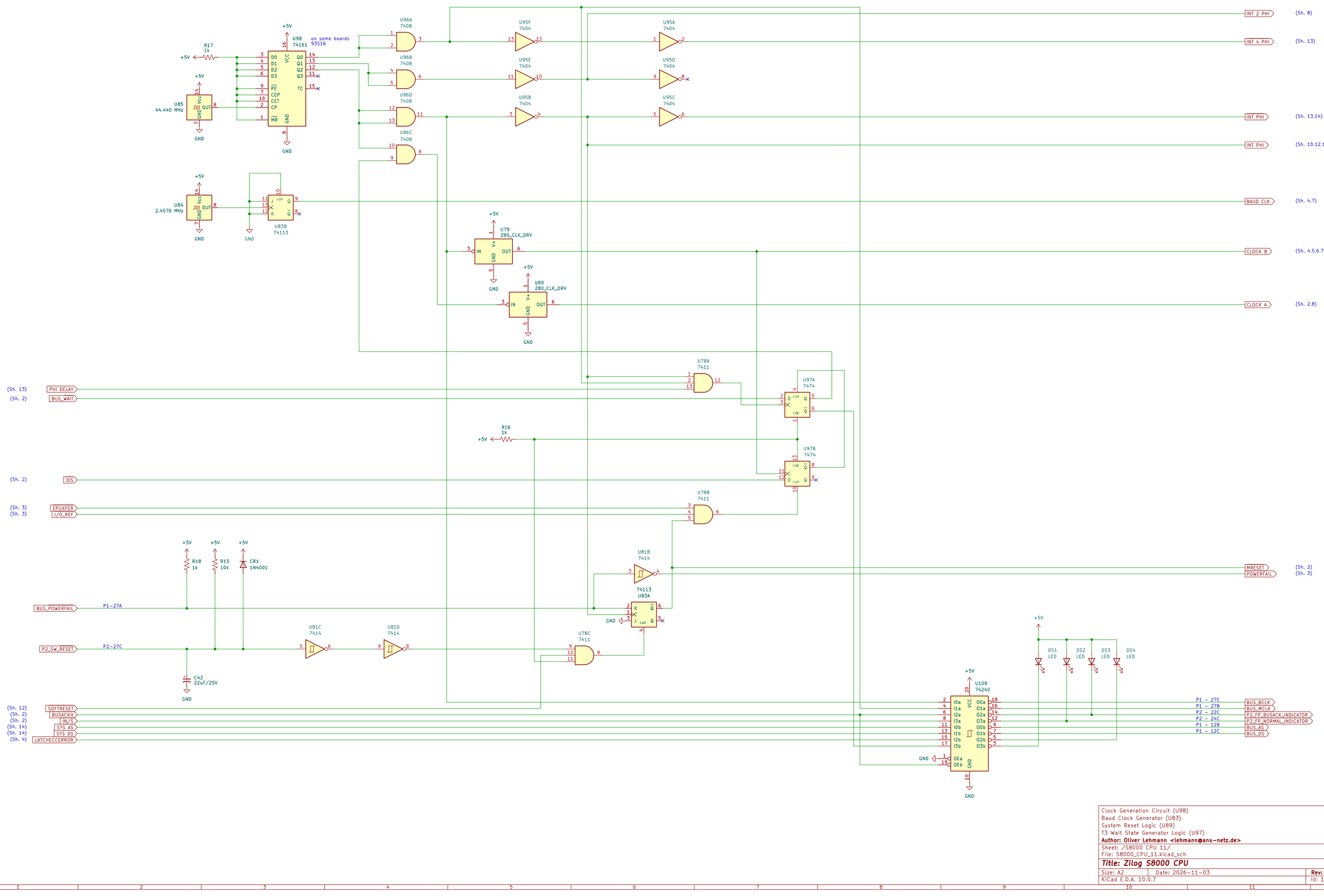
Sheet: /S8000 CPU 09/
File: S8000_CPU_09.kicad_sch

Title: Zilog S8000 CPU

Size: A3	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 9/16

(Sh. 2) ISNAD6
(Sh. 2) ADIN[0..15]
(Sh. 2) LAD[0..15]
(Sh. 2) ISNAD6
(Sh. 2) ISNAD0
(Sh. 14) BDMEMON
(Sh. 2) MREQ
(Sh. 2) MREQ
(Sh. 14) BDMEMON
(Sh. 12) SELNBRKREG
(Sh. 11) INT PHI
(Sh. 2) DS
(Sh. 2) SN[0..6]
(Sh. 2) IW/R
(Sh. 12) SELSYSBAKREG
(Sh. 2) IR/W
(Sh. 14) SEG USER
(Sh. 2) IN/S
(Sh. 14) NONSEGUSER
(Sh. 2) IST2
(Sh. 2) IST3
LAD15
LAD13
LAD14
GND
+5V
U55 74138
U34B 74260
U23C 7410
U3C 7408
U21D 7432
U38 7408
U22D 7432
U38A 7408
U61B 74260
U5B 7474
U22A 7432
U18A 7402
U18C 7402
U71 74373
U72 74373
ADIN5
ADIN7
ADIN2
ADIN0
ADIN3
ADIN1
ADIN4
ADIN6
BRK5
BRK7
BRK2
BRK0
BRK3
BRK1
BRK4
BRK6
SN3
SN5
SN2
SN4
SN1
U73 74244
U53 7485
U52 7485
BRK[0..7]
LAD12
LAD13
LAD14
LAD15
LAD8
LAD9
LAD10
LAD11
DATA REF
1k
R11
U12A 74244
ADIN0
ADIN1
ADIN2
ADIN3
U1D 7404
U29E 7404
U16A 7451
U16B 7451
BDMEMA2K (Sh. 9)
BDMEMB2K (Sh. 9)
BDMEMC2K (Sh. 9)
IMEML (Sh. 12,13,14)
IMEMH (Sh. 14)
MANUAL_NMI (Sh. 3)
POWERFAIL (Sh. 3)
LATCHECCERROR (Sh. 4)
ENAONBDNMI (Sh. 3)
MSTACK (Sh. 8)
MDATA (Sh. 8)
NON SEGMENTED
SEGMENTED
E11
E12
E10
E3
E2
E1
E4
E5
E6
E7
E8
E9
U17C 7400
U15B 7402
U15C 7402
U29C 7404
U30B 7410
U2C 7410
U23A 7410
U29F 7404
U3A 7408
U17A 7400
R1
4.7k
MCODE (Sh. 8)
SEGVIOL (Sh. 8)

Memory Management Control Logic
Parity Error Checking: error buffer (U12)
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 10/
File: S8000_CPU_10.kicad_sch
Title: Zilog S8000 CPU
Size: A2 | Date: 2026-11-03 | **Rev: 1.1**
KiCad E.D.A. 10.0.7 | Id: 10/16

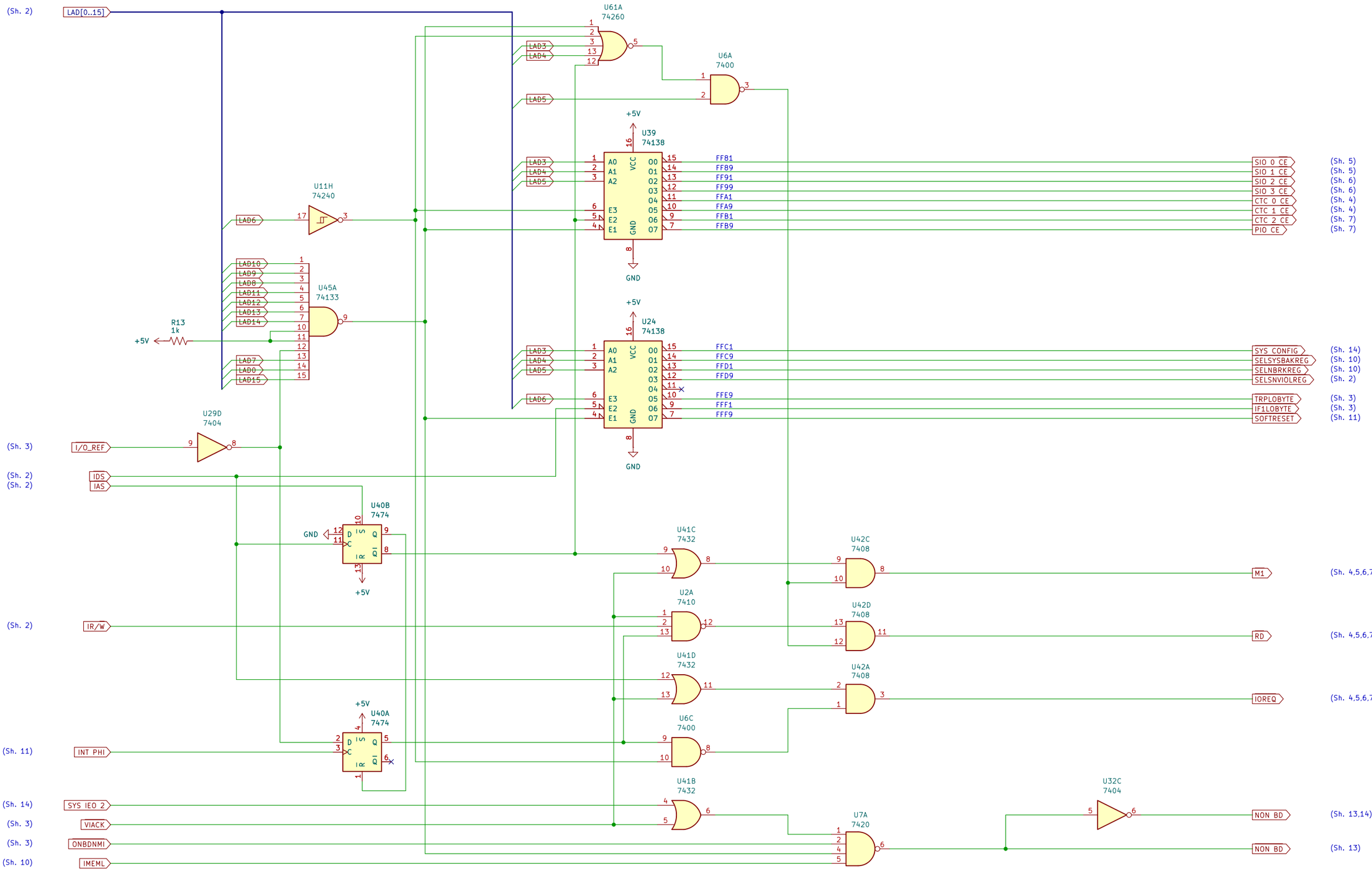


Clock Generation Circuit (U98)
Baud Clock Generator (U83)
System Reset Logic (U89)
T3 Wait State Generator Logic (U97)
Author: Oliver Lehmann <lehmann@ans-netz.de>

Sheet: /S8000 CPU 11/
File: S8000_CPU_11.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 11/16



Peripheral Handshaking Logic (U24, U39)

Author: Oliver Lehmann <lehmann@ans-netz.de>

Sheet: /S8000 CPU 12/

File: S8000_CPU_12.kicad_sch

Title: Zilog S8000 CPU

Size: A3	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 12/16

(Sh. 2) IAS

R7
1k
+5V

U13A
7474

(Sh. 11) INT PHI
(Sh. 11) INT 4 PHI

U13B
7474

GND

PHI DELAY (Sh. 8.11)

ADDR ENAL (Sh. 14)

(Sh. 12) NON BD

U30C
7410

BD_DATAEN (Sh. 2)

(Sh. 4) ENA BYTE SWAP

U31A
7420

(Sh. 2) IDS

U21C
7432

(Sh. 2) BUSACK

(Sh. 3) MMU/EPUREF

U33B
7411

U31B
7420

(Sh. 2) IW/R

(Sh. 2) IDS

U15D
7402

U29B
7404

(Sh. 14) MMUONH
(Sh. 10) IMEML
(Sh. 2) IST3

U23B
7410

U21A
7432

(Sh. 4) ENA BYTE SWAP
(Sh. 12) NON BD

U17D
7400

(Sh. 2) IAD[0..15]

(Sh. 2) LAD[0..15]

(Sh. 2) ISNAD[0..7]

(Sh. 13) TRAD[8..23]

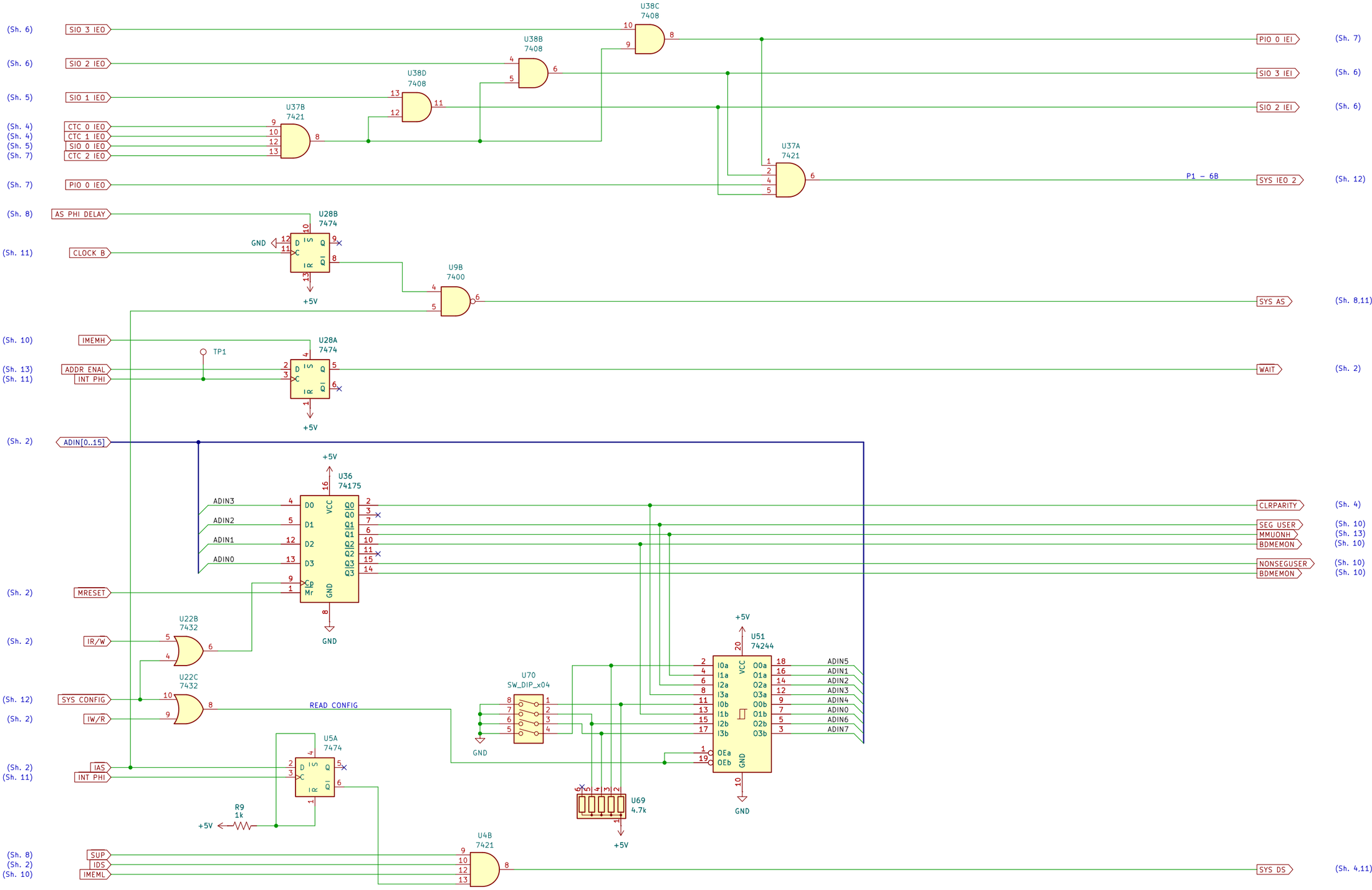
U114 74244
U117 74244
U116 74244
U115 74244
U110 74244
U113 74245
U111 74245
U112 74244

BUS_AD[0..23]

Address/Data Buffers and Steering Logic
Byte Swap Buffer (U112)
Author: Oliver Lehmann <lehmann@ans-netz.de>
Sheet: /S8000 CPU 13/
File: S8000_CPU_13.kicad_sch

Title: Zilog S8000 CPU

Size: A2	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 13/16

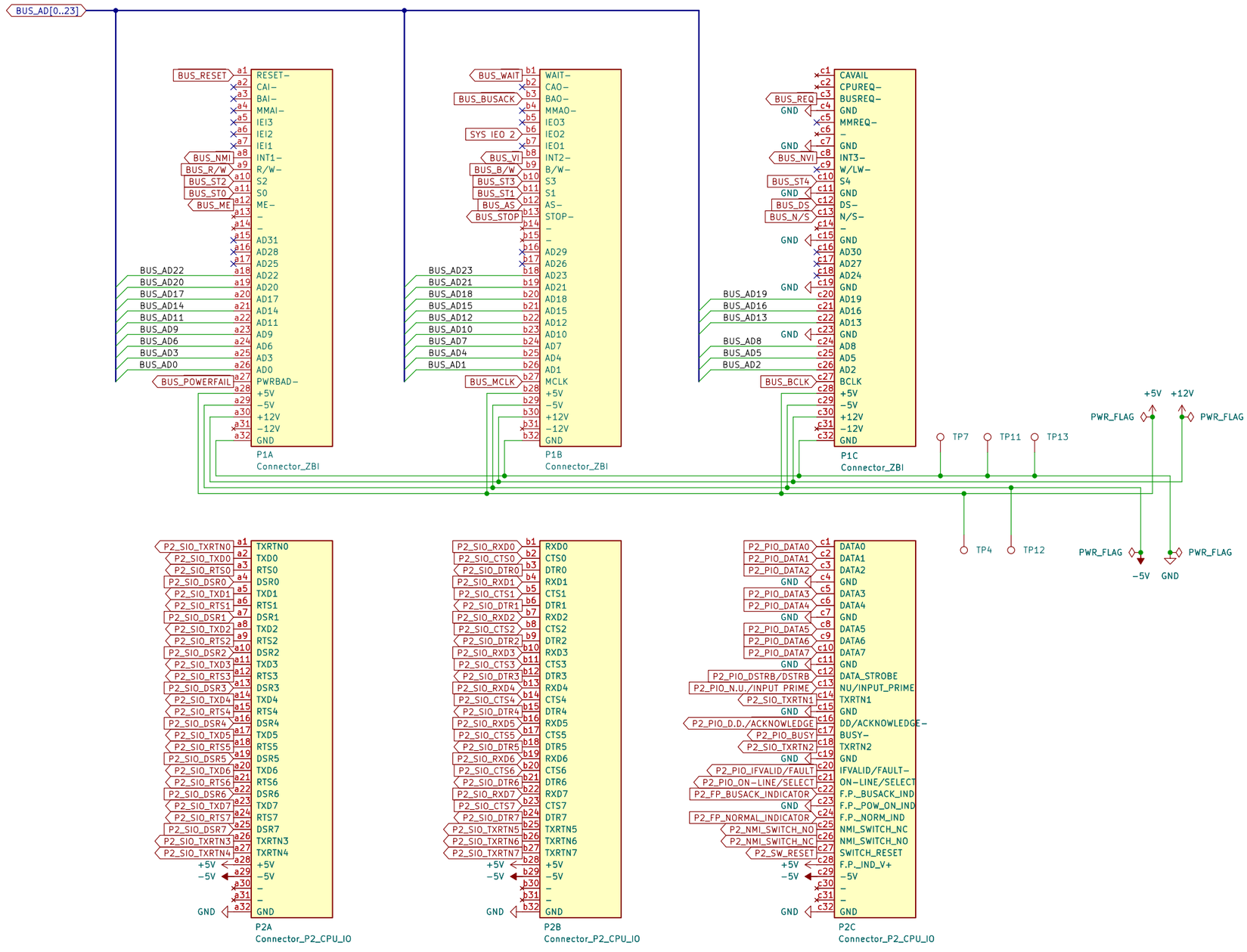


Interrupt Priority Connection
System Configuration Register
T2 Wait State Generator Logic (U28)
Author: Oliver Lehmann <lehmann@ans-netz.de>

Sheet: /S8000 CPU 14/
File: S8000_CPU_14.kicad_sch

Title: Zilog S8000 CPU

Size: A3	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 14/16



Connectors

Author: Oliver Lehmann <lehmann@ans-netz.de>

Sheet: /S8000 CPU 15/

File: S8000_CPU_15.kicad_sch

Title: Zilog S8000 CPU

Size: A3	Date: 2026-11-03	**Rev: 1.1**
KiCad E.D.A. 10.0.7		Id: 15/16